



LoRD: LoRA-Adapted Re-Identification with DINO

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Problem

- Video Person Re-ID**: retrieve gallery tracklets of the same person as a query tracklet across disjoint cameras.
- Tracklets carry occlusion, detector noise, viewpoint change and redundant frames. **Temporal attention may help pick the right frames**.
- DINO gives strong general features, but **frozen features are not calibrated for identity matching**.

input query tracklet $T \times 3 \times H \times W$

output ranked gallery tracklets

metrics mAP · CMC Rank-1 / 5 / 10

Dataset: MARS

1,191

identities
625 train / 626 test

8,298

training tracklets
across 6 cameras

9,330

gallery tracklets
(3,248 distractors)

Frames resized to 252×126 for pedestrian aspect & ViT-L/14 patch alignment. Train samples $T=4$; eval at uniform $T=4/8/12$.

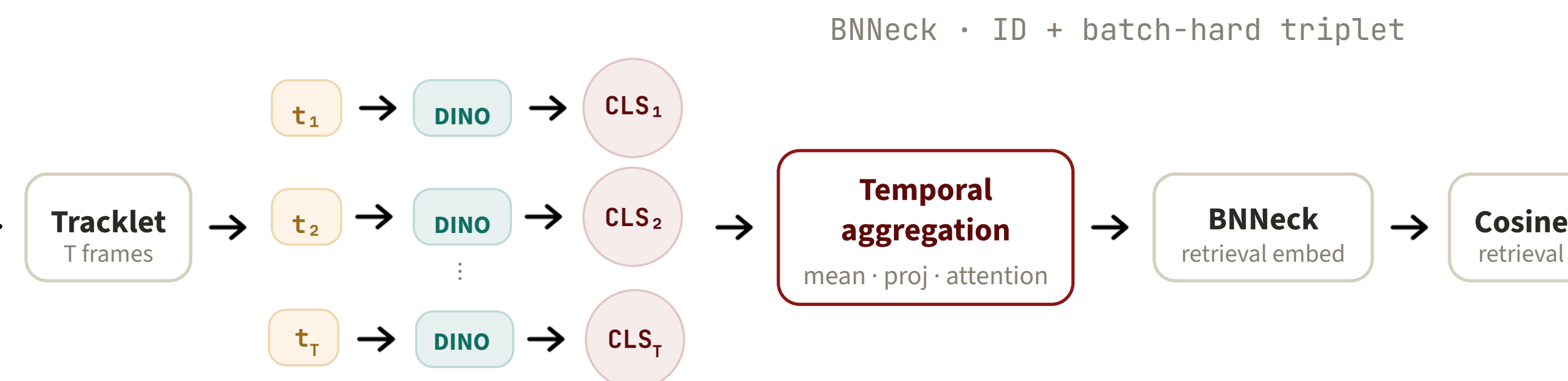
Tracklet-consistent padded crop, h-flip, random erasing; final LoRA runs use **PK sampling** ($P=8, K=4$).

Study Design

What drives video re-ID with DINO — **backbone adaptation**, **temporal aggregation**, or both?

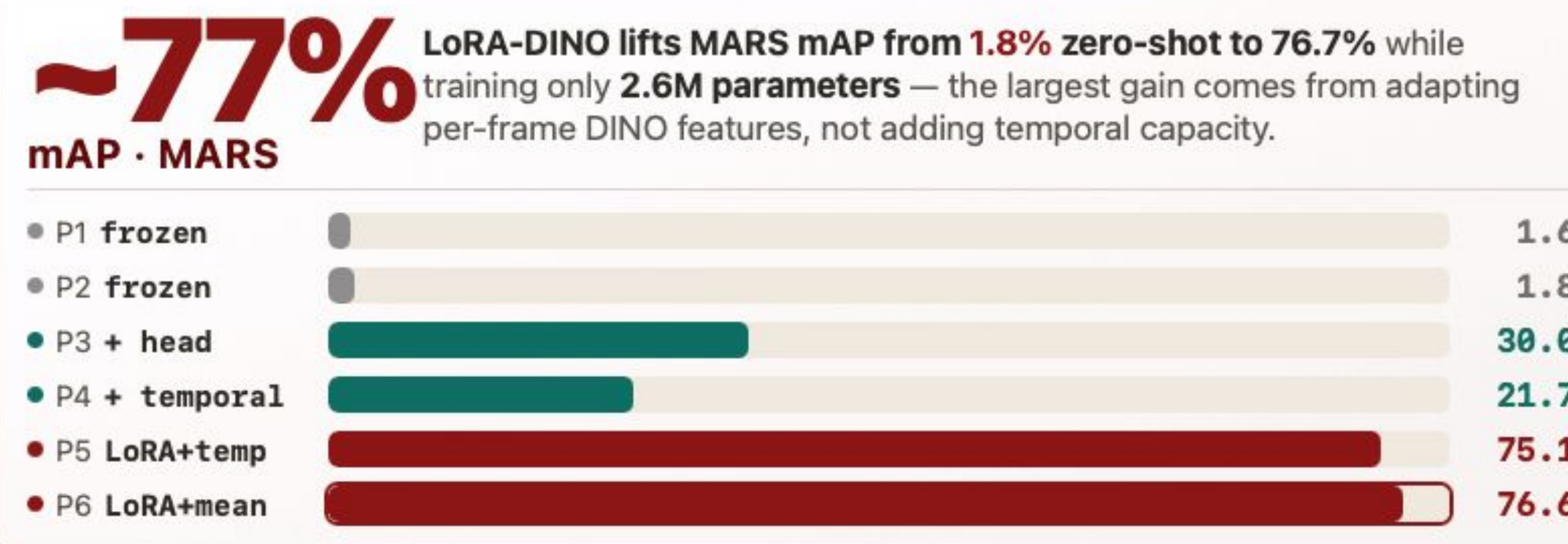
P1	DINOv3 L/16	frozen	mean	zero-shot
P2	DINOv2 L/14	frozen	mean	zero-shot
P3	DINOv2	frozen	mean + proj	learned head
P4	DINOv2	frozen	transformer	temporal only
P5	DINOv2	LoRA	transformer	adapt + temp.
P6	DINOv2	LoRA	mean	adapt only

Model



Frames are encoded **independently** by frozen or LoRA-adapted DINO. **CLS tokens** are aggregated by mean pooling, mean + projection, or a temporal transformer; BNNeck trains identity-discriminative embeddings for cosine retrieval.

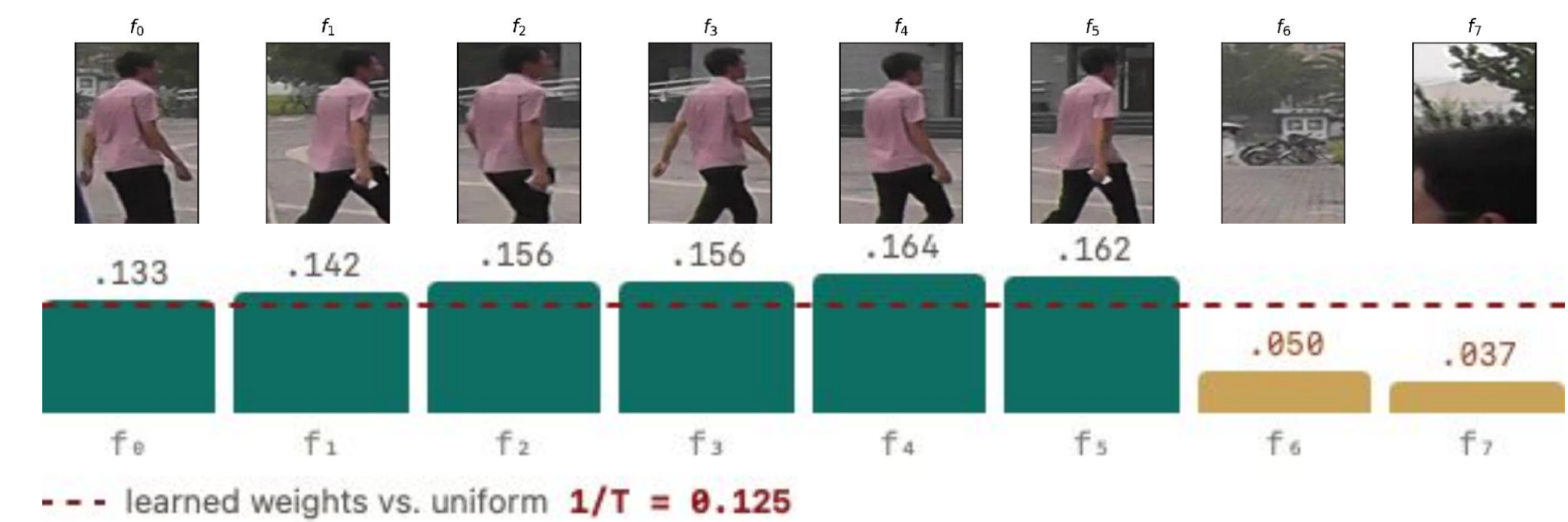
LoRA $r=8$, $a=16$, MLP layers temporal: 1 block · 8 heads zero-init residual $\approx$ mean
triplet margin 0.3 label smooth 0.1



Main Ablation Results

MODEL	TRAINABLE	T	MAP	R-1	R-5
P1 · DINOv3 zero-shot	0	8	1.6	3.8	7.6
P2 · DINOv2 zero-shot	0	8	1.8	3.6	8.0
P3 · frozen + mean/proj	1.7M	8	30.0	40.5	62.1
P4 · frozen + temporal	13.3M	4	21.7	31.7	52.1
P5 · LoRA + temporal	15.2M	8	75.1	78.7	89.5
P6 · LoRA + mean	2.6M	8	76.7	80.3	89.8

Why Attention Doesn't Work



99.7%

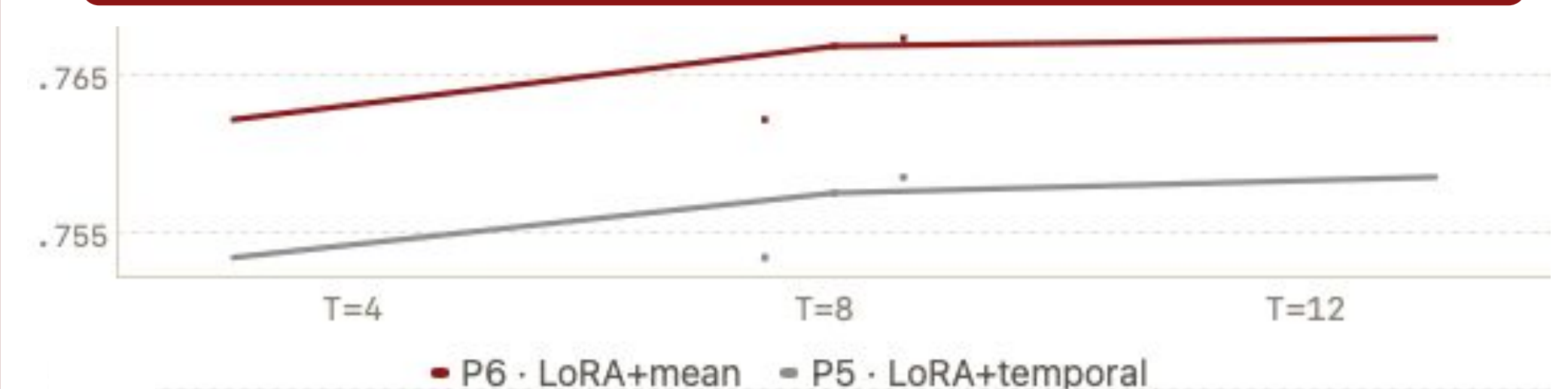
of max pooling entropy
(avg over 1,980 queries)

0.97–0.99

cosine sim of adapted frame CLS
in a tracklet

The head mildly down-weights off-target frames (f_6, f_7), but final pooling stays near-uniform. Since adapted frame embeddings already align, weighted pooling $\approx$ mean pooling.

Frame Count Sensitivity



More inference frames give small monotonic gains (<1 mAP total). **P6 stays ahead of P5 at every T** — extra frames don't unlock temporal value.

Context, Not SOTA

METHOD	FAMILY	R-1	MAP
Gao & Nevatia (attn)	CNN+temporal	83.3	76.7
AP3D	3D CNN	90.1	85.1
TF-CLIP	CLIP video	93.0	89.4
KeyRe-ID	ViT video	97.3	91.7
LoRD P6 (direct)	LoRA-DINO	~80	~77

Published systems use task-specific architectures, heavier recipes, or re-ranking. P6 is a strong **direct-retrieval baseline at 2.6M params**.

Takeaways

- Backbone adaptation is the driver.** LoRA on DINOv2 MLP layers gives the largest jump.
- Temporal attention is redundant.** P5 adds 12.6M params over P6 yet underperforms mean pooling.
- Frame features already agree.** Adapted per-frame CLS embeddings are highly similar within a tracklet.

- Frame count isn't the bottleneck.** $T=4/8/12$ differ by <1 mAP.
- Simple is competitive.** LoRA-DINO mean pooling reaches $\sim 77\%$ mAP via direct retrieval at 2.6M params.

FUTURE WORK

Sweep LoRA rank $r \in \{4, 8, 16\}$; MLP vs Q/V placement
Patch-token / multi-token pooling beyond CLS-only
k-reciprocal re-ranking as post-processing
Transfer to LS-VID, MEVID, occluded video re-ID

REFERENCES

MARS— Zheng et al., ECCV 2016.
DINOv2— Oquab et al., 2023.
LoRA— Hu et al., ICLR 2022.